

MLOps–based Data Science Process Model

General Information and Business Understanding

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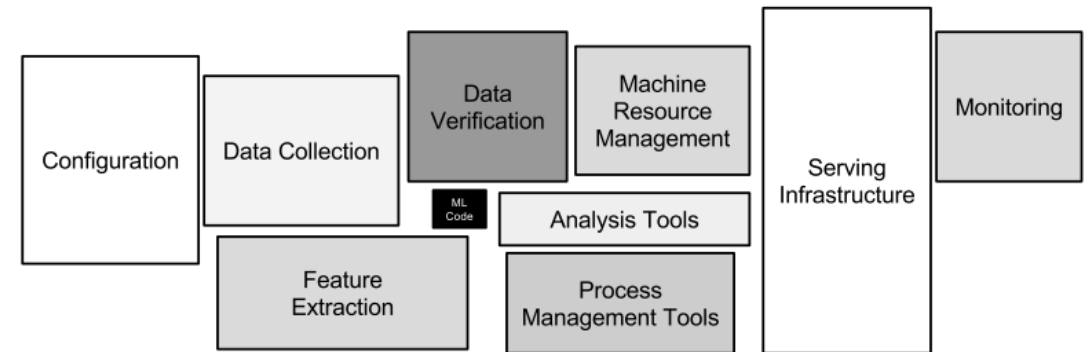
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Scientific Background: Data Science and Process Models

- Data Science (DS) projects have become essential to modern enterprises (Müller et al. 2018), generating insights that support decision-making, reduce costs, and improve business processes (Saltz & Krasteva 2022)
 - Data Science: interdisciplinary field aiming to extract “useful knowledge directly from data through a process of discovery or of hypothesis formulation and hypothesis testing” (Chang & Grady 2019)
 - Frequently involves Machine Learning (ML), which includes algorithms that learn from training data to uncover insights or patterns without explicit programming (Bishop 2006; Janiesch et al. 2021)
- DS initiatives, particularly due to ML, are complex, involving various technical and managerial challenges → robust project management framework required
 - DS process models (DSPMs): approaches, processes, or frameworks encompassing a range of stages, activities, and techniques (Rösl & Schieder 2024) to enable structured and efficient DS project management (Haertel et al. 2022, Rösl et al. 2025)

Scientific Background: System Architecture Perspective

- DS/ML offer potential to build automated and complex prediction systems for various use cases
- Prioritizing rapid and cheap development and deployment can lead to difficulties in ML system maintenance → technical debt (Sculley et al. 2015)
 - Technical debt is more emphasized in ML systems compared to traditional software
 - Data dependencies
 - Feedback loops
 - ML code smells
 - Configuration debt
 - Changes in the external world
 - ...
- Approaches need to reduce technical debt



Only a small portion of ML systems consists of ML code (Sculley et al. 2015)

Scientific Background: MLOps

- Difficulties and „technical debt“ (Sculley et al. 2015) in developing, deploying, and maintaining ML-based products
- MLOps (ML Operations): extends DevOps principles to DS domain to help create more efficient and robust ML systems (Kreuzberger et al. 2023)

“MLOps (Machine Learning Operations) is a paradigm, including aspects like best practices, sets of concepts, as well as a development culture when it comes to the end-to-end conceptualization, implementation, monitoring, deployment, and scalability of machine learning products. Most of all, it is an engineering practice that leverages three contributing disciplines: machine learning, software engineering (especially DevOps), and data engineering.” (Kreuzberger et al. 2023)

- MLOps principles (Kreuzberger et al. 2023)
 - CI/CD automation; workflow orchestration; reproducibility; versioning of data, model, code; collaboration; continuous ML training & evaluation; ML metadata tracking, continuous monitoring, feedback loops

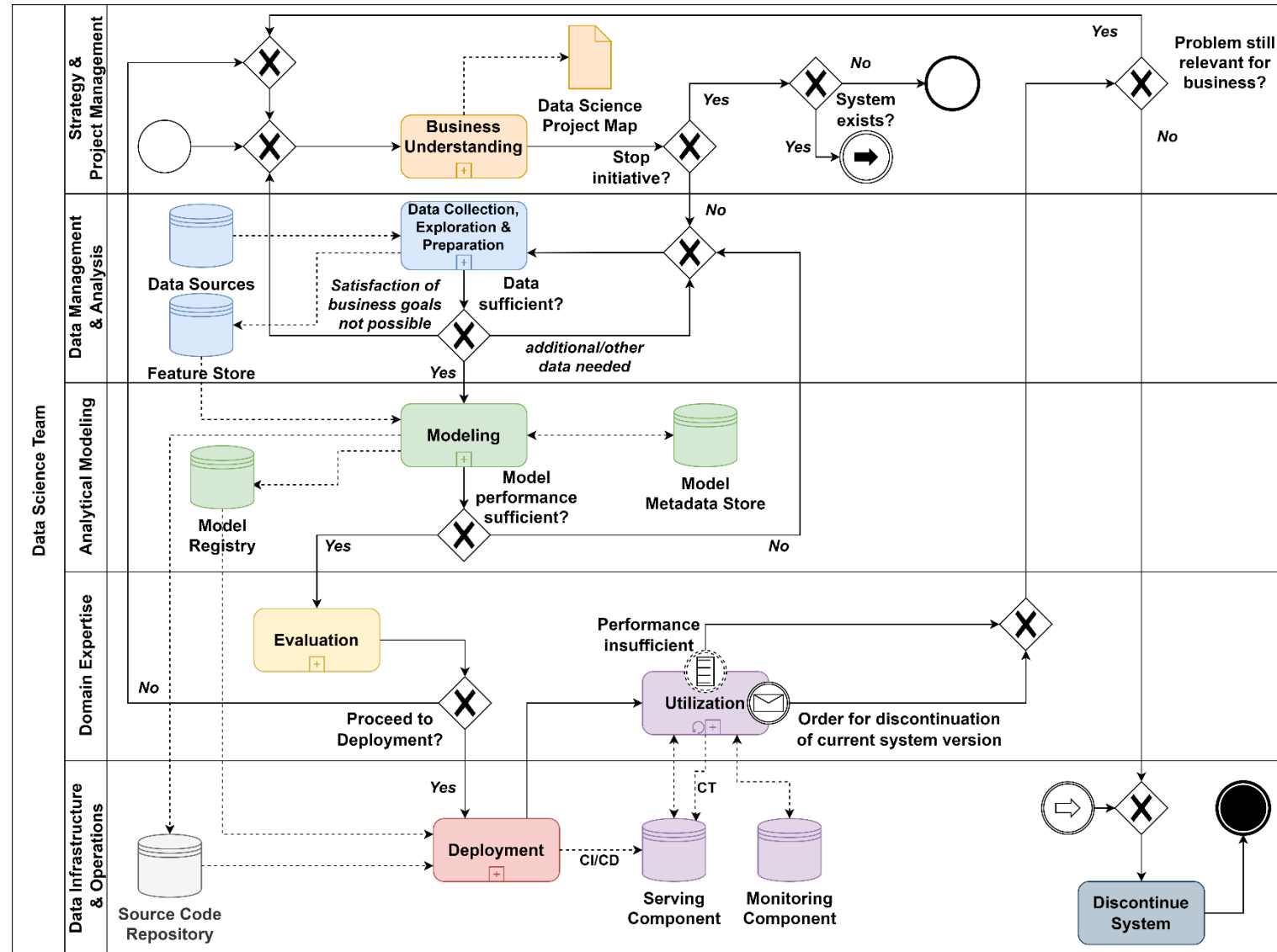
MLOps–Based Data Science Process Model

MLOps–DSPM

- Why?
 - The realization of data-intensive, analytical initiatives is facilitated with a process model guiding project execution (Kutzias et al. 2023)
 - Data Science projects often involve Machine Learning, whose difficulties in model deployment and development can be addressed with MLOps.
- Structure
 - 14 formalized diagrams using BPMN (Business Process Model and Notation) and UML (Unified Modeling Language) models
 - two models (one BPMN + one UML) covering abstract project perspective
 - two models for each of the six Data Science project lifecycle stages
 - Best practices and instructions for each model/step

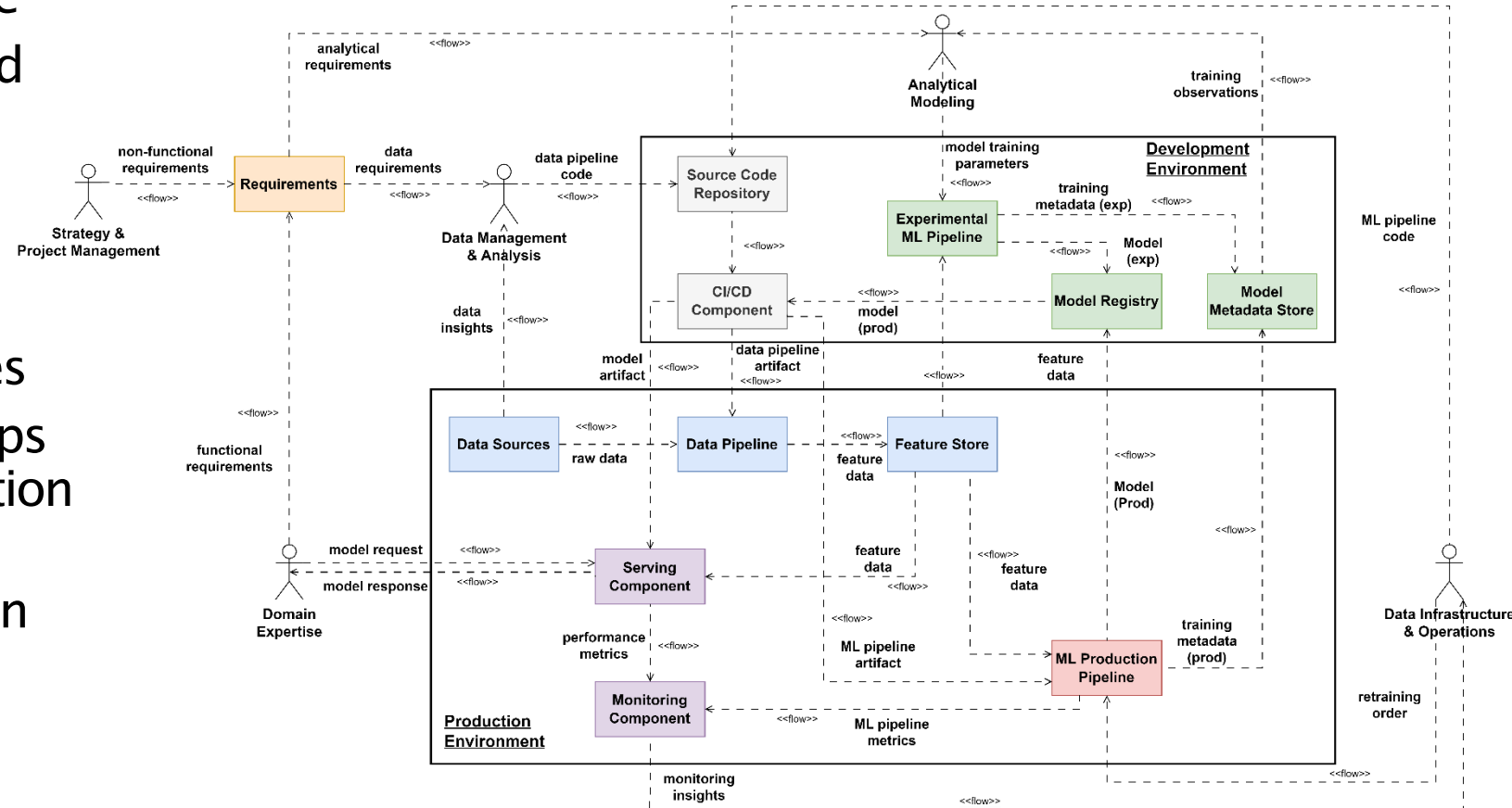
An MLOps-based Data Science Process Model – Abstract Process View

- BPMN role
 - Process view of the DS project → flow of tasks and decisions
- Components
 - Tasks: project stages or activities
 - Pools/swimlanes: abstract team roles
 - Data objects: documentation artifacts, data stores, or MLOps components
- BPMN limitation
 - cannot model the interaction between data objects



Abstract Information Flow View

- UML information flow role
 - Model the information and data flow between actors, MLOps components, and documentation artifacts
- Components
 - Actors: abstract team roles
 - Information objects: MLOps components, documentation artifacts
 - Flow: data and information flow
- UML information flow limitation
 - Cannot model sequential and logical order of tasks



Key MLOps Components (1 / 3)

- Feature Store
 - centralized repository designed for managing, storing, and serving ML features for experimentation and serving
- CI/CD component
 - Ensures build, test, delivery, and deploy steps for data/ML pipelines
 - Allows for rapid feedback to developers
 - Includes the execution of various tests (e.g., unit, integration tests)
- Source Code repository
 - training, inference, and application source code are versioned in a repository
 - fosters collaboration
- Data pipeline
 - Build on the insights from exploratory data analysis
 - Extracts data from the sources and performs automated cleaning and transformation steps
 - Loads data into feature store system

Key MLOps Features and Components (2/3)

- Experimental ML pipeline
 - Used by data scientists to estimate best combination of algorithms and hyperparameters
 - Iterative process
- ML Production pipeline
 - Build based on experimentation insights
 - Extracts data from feature store, performs data split, and trains a model (exported to model registry)
- Model Registry and Metadata Store
 - Stores trained ML models and their metadata → reproducibility
 - Tracking and logging of metadata for each training job (experimental and productive ML pipelines)

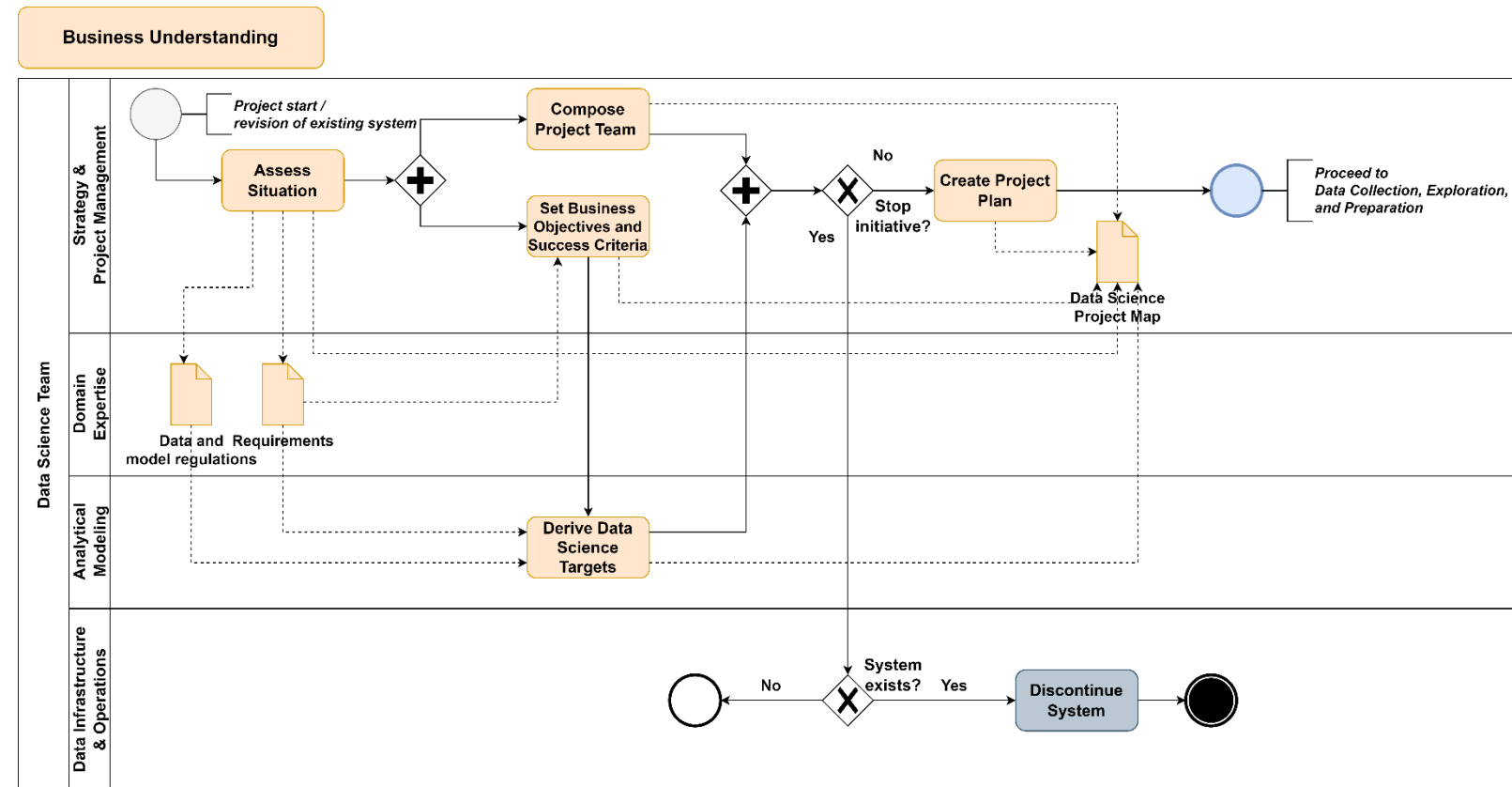
Key MLOps Features and Components (3/3)

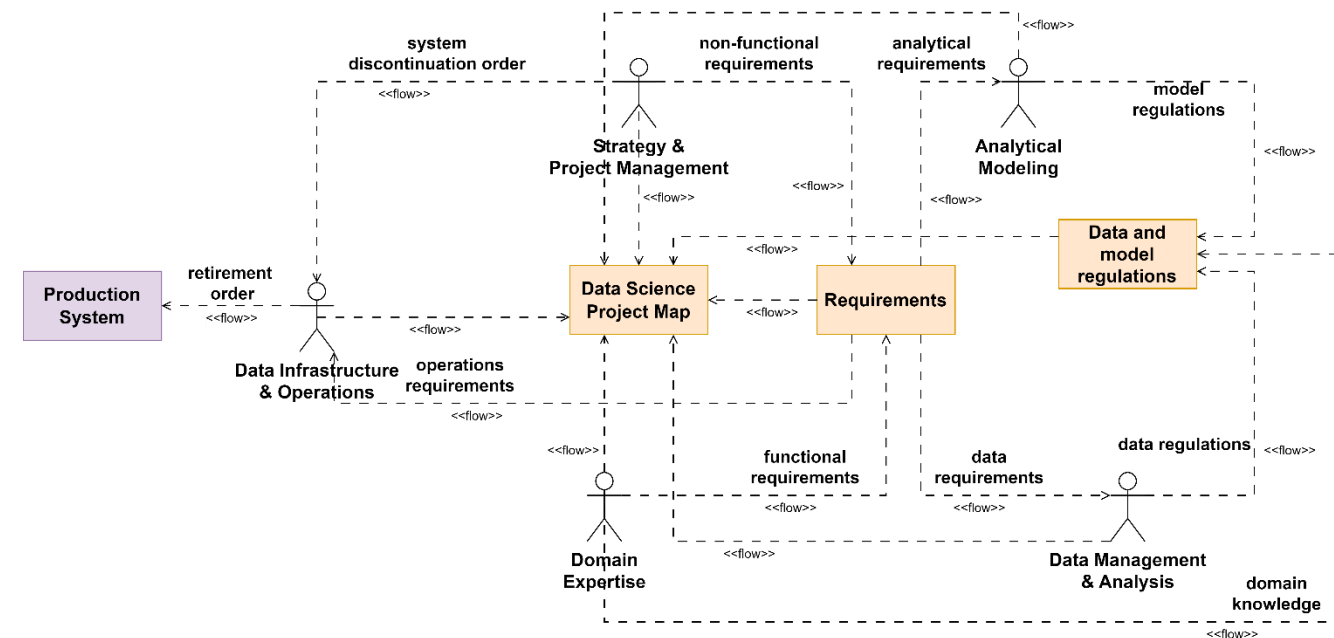
- **Serving component**
 - Deployed model makes predictions on new, unseen data
 - Implementation depends on use case
- **Monitoring component**
 - continuous assessment of data, model, code, and infrastructure
 - Can trigger periodic retraining of the model based on new data, requires automated ML workflow pipeline
 - Retraining requires careful consideration of costs
- **Environment separation**
 - DS/ML explorative in nature → require experimentation
 - Separation of experimental and productive environments
 - Deployment to productive environments streamlined via CI/CD component

Business Understanding

Business Understanding – Process Perspective

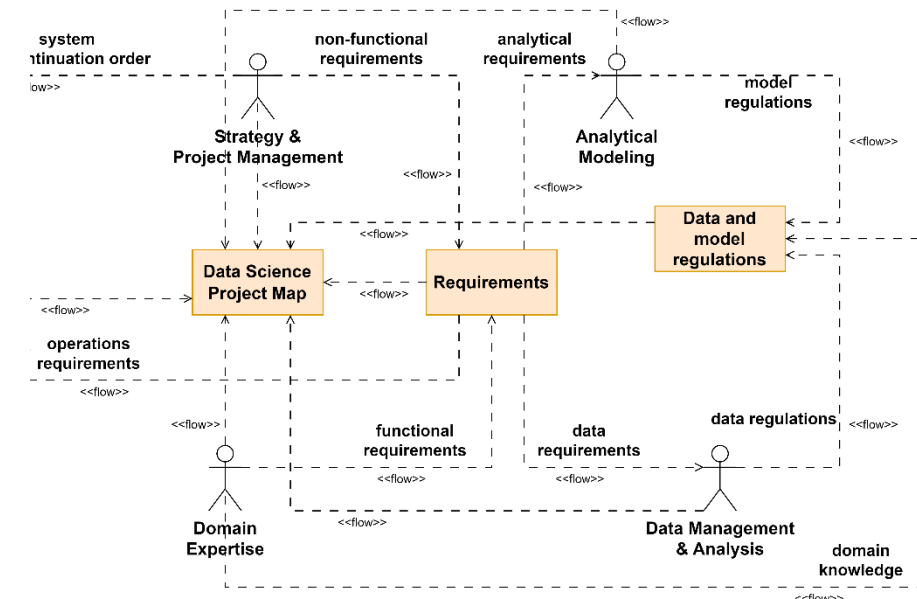
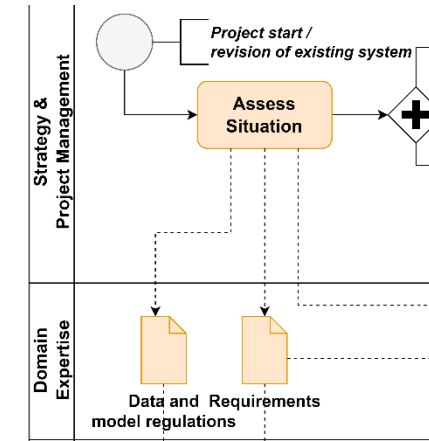
- Conceptualization and planning of the project
 - Entry point of the initiative or revisitation of existing solution
- Main tasks
 - Assess situation
 - Compose Project Team
 - Set Business Objectives and Success Criteria
 - Derive Data Science Targets
 - Create Project Plan
- Artifacts
 - Data and model regulations
 - Requirements
 - Data Science project map





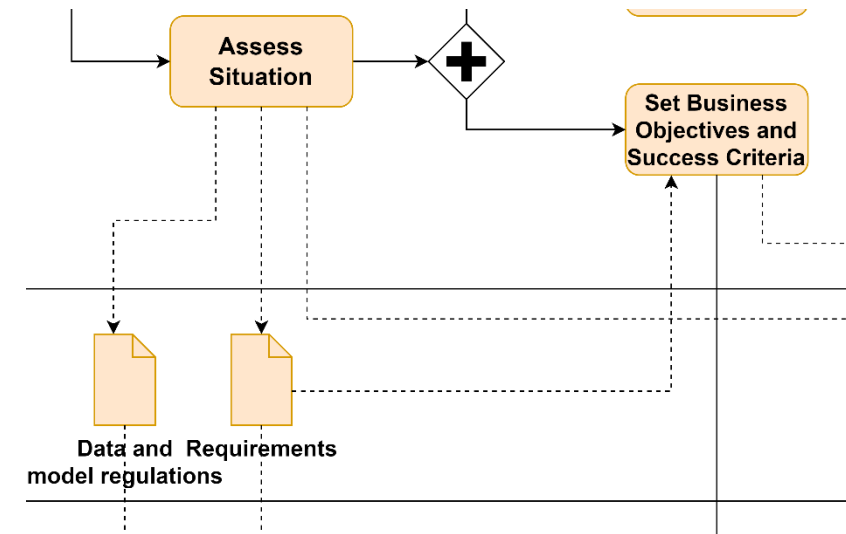
Assess Situation

- Goal: analysis of the problem, including its suitability, and definition of requirements
 - Can rely on knowledge and best practices from previous initiatives
 - Identify risks and required resources
- Various requirements from different stakeholders
 - Functional and non-functional requirements
 - Operations requirements
 - Analytical requirements
 - Data requirements
- Artifact output
 - Data and model regulations
 - Requirements



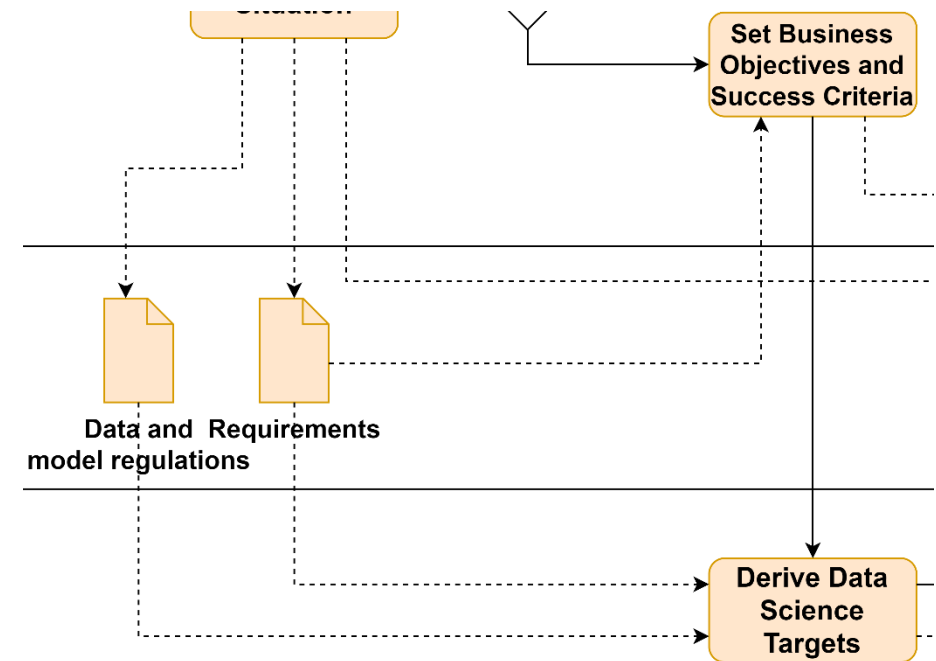
Set Business Objectives and Success Criteria

- Goal: identify objectives and associated success metrics from the business perspective for the project
 - Avoiding answering the wrong questions (Chapman et al. 2000)
- Definition of objectives should follow the SMART principle
 - Specific, measurable, achievable, relevant, time-bound
- Success metrics (KPIs) indicate fulfillment of a business objective
- Different levels of objectives can be visualized in a goal hierarchy
- Influenced by desired form of final product



Derive Data Science Targets

- Goal: identify Data Science objectives and metrics
 - Derived from the business objective(s)
- Requires definition of analytical problem type during situation assessment
 - Influenced by potential model and data regulations (e.g., explainability, protection of certain features)
- Depending on problem type, the use of multiple metrics might be sensible
 - Example: classification tasks on imbalanced datasets

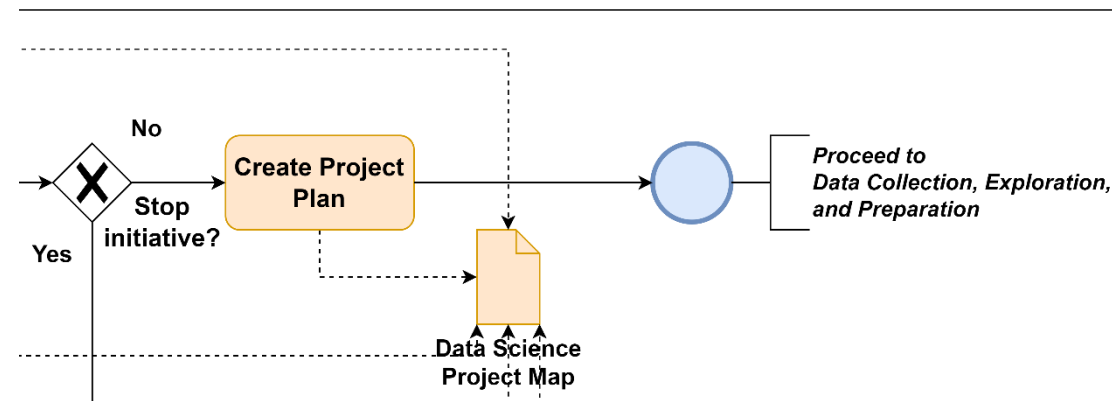


Compose Project Team

- Goal: compose interdisciplinary team covering all relevant knowledge areas and competencies
 - Define roles and responsibilities (prescribed on a high-level by the process model)
 - Team coordination and collaboration strategies
- Abstract Data Science team role categories (Haertel et al. 2025)
 - Domain Expertise: business process knowledge and domain data structure
 - Strategy & Project Management: project guidance and management aspects
 - Data Management & Analysis: data access, collection, exploration, and preparation
 - Analytical Modeling: building and evaluating (ML) models
 - Data Infrastructure & Operations: responsible for technological infrastructure for model deployment, maintenance, and scaling

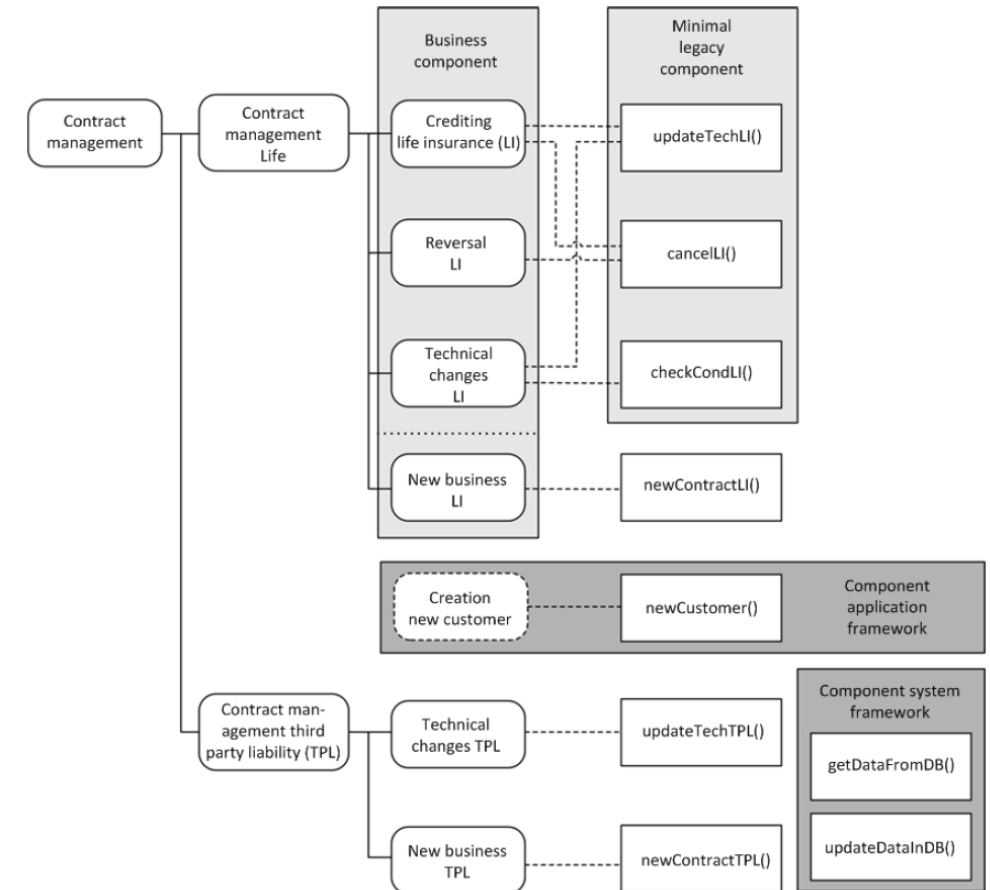
Create Project Plan

- Goal: preliminary planning of the project
 - Fundamental prerequisite for project continuation
 - Extends to managerial and technical aspects
- Shaped by the results of the previous Business Understanding activities
- Can rely on key tools of project management
 - Milestone plan
 - Cost/benefit analysis
 - Requirements catalogue
 - Risk identification, assessment, and mitigation strategies
 - Glossary
 - ...



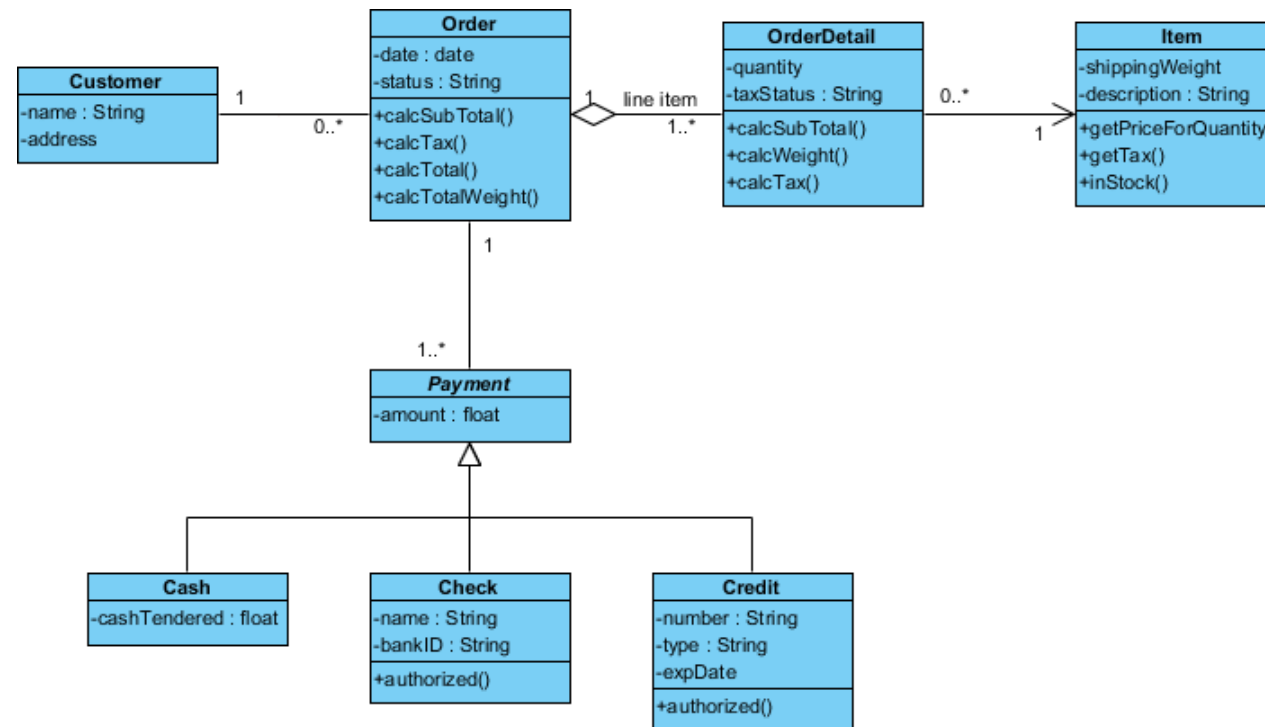
Create Project Plan

- Technical planning (system architecture perspective)
 - ML-based systems rarely exist in isolation
→ embedded in the overall business technology landscape
- Business component identification
 - Functional partition of application system to be designed → functional decomposition diagram for representing business processes, functions, and elementary functions → mapping of methods to elementary functions



Create Project Plan

- Implementation of components requires specification of data to be used and processed → data model
- Can be modelled using UML class diagrams



Data model example for a webshop, source: <https://dataedo.com/blog/basic-data-modeling-techniques>

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